Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/24/2024
PI Name:	Clement, Gilles Ph.D.		
Project Title:	Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) (PI: Clement)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3711	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/30/2019	**End Date:**	09/30/2033
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Project start date changed 01/30/2019 to match original investigation, as requested by NASA HHC (Ed., 1/3/25). NOTE: End date changed to 09/30/2033 per C. Ribeiro/NASA-JSC. (Ed. 10/5/23). NOTE: Project start date updated due to P.I. change to G. Clement. Information per C. Ribeiro/NASA-JSC. (Ed. 8/29/22).		
Key Personnel Changes/Previous PI:	September 2024: Dr. Clement will be retiring from NASA. For information on the continued investigation, see: "Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) (PI: Macaulay)": https:// . Beginning in 2024, the Principal Investigator will be Timothy Macaulay, Ph.D. Timothy Macaulay, PhD, has been added as a co-Investigator. He will be primarily responsible for conducting pre- and post-flight testing, as well as assisting ROI in Mission Control Center for off-nominal situations during in-flight testing		
COI Name (Institution):	Dervay, Joseph M.D. (NASA Johnson Space Center) Makishima, Tomoko M.D., Ph.D. (University of Texas Medical Branch at Galveston) Wood, Scott Ph.D. (NASA Johnson Space Center) Macaulay, Timothy (KBR)		

Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Adaptation to the absence of Earth's gravitational environment during spaceflight causes neurological disturbances that are either directly or indirectly mediated by the vestibular system. These disturbances include space motion sickness, spatial disorientation, and cognitive impairment, as well as changes in head-eye coordination, vestibulo-ocular reflex, and interactions with support surfaces. After return to Earth, astronauts experience other vestibular-driven behavioral changes, including re-entry motion sickness, motion illusions, gaze-induced nystagmus, and balance and locomotion deficits. Otolith-mediated reflex gain changes are striking shortly after g-transitions. However, animal studies have shown that structural modifications of the vestibular sensory apparatus may occur throughout an extended spaceflight exposure. To date, no long-duration flight studies have directly investigated potential changes in the vestibular organs of astronauts. As a part of the Complement of Integrated Protocols for Human Research (CIPHER, a set of 14 studies sponsored by NASA and international partner agencies, https:// this study will address this gap by performing a systematic neuro-vestibular examination of crewmembers in orbit at regular intervals, as well as immediately after landing. Crewmembers from short-duration, six-month, and one-year missions will be recruited to investigate temporal changes, and to identify trends in adaptation to vestibular health and performance. In orbit, subjects will perform active movements and be exposed to various passive maneuvers executed by the operator. Observations and recordings of eye, head, and body movements, as well as subjective perception of motion and verbal reports, will be used for evaluating the presence of abnormal eye movements, dysmetria, motion sickness symptoms, and illusions of motion during head or body movements. Tests will be performed both early in the mission and once every one or two months thereafter. The post-flight examination will be performed on R+0 and several times thereafter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The tests proposed in this study are well established and validated protocols that are able to detect acute or chronic vestibular syndromes. Repeated measurements during adaptation to g-transitions will provide insight into the vestibular compensation process. If the observed physiological changes in the crewmembers are more deleterious after the year-long International Space Station (ISS) expeditions than those documented after standard-duration ISS expeditions, then relevant countermeasures will be required to enable longer duration missions. Depending on the etiology of the vestibular disorders revealed by our tests, monitoring for long-term health outcomes and vestibular rehabilitation countermeasures can be tailored to the deficits observed
Task Progress:	Ed. Note - September 2024: Dr. Clement will be retiring from NASA. For additional information on this investigation, see: 1) Prior Reporting: "Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) (PI: Reschke)": https://. From 2019-2022, the Principal Investigator was Millard Reschke, Ph.D. 2) Current Reporting: "Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) (PI: Macaulay)": https://. Beginning in 2024, the Principal Investigator will be Timothy Macaulay, Ph.D. ---------- This past reporting period, the Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) study continued collecting data as part of the Complement of Integrated Protocols for Human Exploration (CIPHER). [Ed. Note: CIPHER is the short title for a set of 14 studies sponsored by NASA and international partner agencies (https://). The first six inflight data collections and the first eight postflight data collections were completed. These major milestones were accomplished despite the challenges of spaceflight operations. For example, on the day of return (R+0), the teams successfully navigated last-minute changes to the landing schedules/locations and used contingency protocols to collect data (average ~8 hours after landing) amid re-entry motion sickness symptoms. To date, two crewmembers have completed all preflight, inflight, and postflight testing, one crewmember is currently undergoing inflight testing, and two crewmembers are currently undergoing preflight testing. The latest Informed Crew Briefings (ICB) were presented to Crew-10 and 73S. Several inflight operator training sessions have also been completed to prepare the crew to help collect these data inflight. The inflight and post-flight motion sickness questionnaires were further customized to support collaboration across related studies. During the latest inflight exam, additional noise was noted in the eye tracking data collected via the commercial binocular video eye tracker on the International Space Station. Our team has worked with the manufacturer to troubleshoot and replicate this issue on the ground. Crew time is being scheduled to conduct a hardware diagnostic test for further inflight troubleshooting. Postprocessing workarounds have been developed to reduce the impact of this noise; thus, the issue will not hinder the collection of these important data. A ground-based study was conducted to compare the balance control and cognitive responses of subjects with bilateral vestibulopathy (BVP) to those of astronauts immediately after they return from long-duration spaceflight aboard the International Space Station. The astronaut data came from a previous study. Thirty subjects with BVP performed five tests using the same procedures as twenty-eight astronauts: sit-to-stand, walk-and-turn, tandem walk, duration judgment, and reaction time. Compared to the astronauts' preflight responses, the BVP subjects' responses were impaired in all five tests. However, the BVP subjects' performance during the walk-and-turn and the tandem walk tests were comparable to the astronauts' performance on the day they returned from space. Moreover, the BVP subjects' time perception and

	reaction time were comparable to those of the astronauts during spaceflight. The BVP subjects performed the sit-to-stand test at a level that fell between the astronauts' performance on the day of landing and 1 day later. These results indicate that the alterations in dynamic balance control, time perception, and reaction time that astronauts experience after spaceflight are likely driven by central vestibular adaptations. Vestibular and somatosensory training in orbit and vestibular rehabilitation after spaceflight could be effective countermeasures for mitigating these post-flight performance decrements. The results of this study were published in a peer-reviewed article. Another ground-based study was conducted to examine variations of the video head impulse test (vHIT), which is conducted as part of the inflight protocol to examine changes in the horizontal vestibulo-ocular reflex in response to high-velocity head movements to detect changes in peripheral vestibular function. Correct interpretation of the flight study data will require consideration of potential artifacts and the constraints of conducting this test across different phases of the mission. Therefore, the aims of our ground study were to (1) examine reliability across different test operators, and (2) compare the results of active (aHIT) versus passive (pHIT) approaches to evaluate the feasibility of self-administered versus operator-assisted approaches. A comparison of responses with visual viewing of a wall target (default condition) versus vision occluded evaluated the influence of other oculomotor control influence across conditions, and a comparison with computer-generated rotator head impulse tests (rHIT) examined the variability associated with the ocular responses independent of the variability in executing the head movements themselves. Seventeen non-astronaut volunteers (male n=12, age=22.9 ± 3.0; female n=5, age=27.0 ± 6.0, mean ± std) completed HIT testing using video-oculography (VOG) goggles and a high-torque rotator system. The test order was counterbalanced across conditions: (1) passive head-on-torso (pHIT, default condition for flight study) using two operators, (2) active head-on-torso (aHIT, subject initiated), and (3) passive head and torso using rotary chair (rHIT). Eye and head movement data were processed to obtain gain in each direction. While the pHIT gains were similar with vision and vision occluded conditions, the percentage of acceptable trials was greater with vision (~93%) versus occluded (~69%). The reliability between operators was acceptable for pHIT gain (Intraclass Correlation, ICC = 0.66, p = 0.001). The percentage of acceptable trials reduced by ~20% during the self-administered aHITs for both visual conditions and tended to be lower than computer-generated rHITs (76% versus 83%). While the aHIT gains were not significantly different than pHIT gains for vision or vision occluded conditions, these measures were poorly correlated. Our findings support the feasibility of using the standard pHIT methodology for spaceflight. Similarities between visual conditions reflect that these responses are mediated by the peripheral lateral canals rather than other oculomotor mechanisms. The inter-tester reliability was acceptable despite differences in the tester training. In addition to concerns about non-vestibular mechanisms introduced during self-administered aHITs, operator-assisted pHITs resulted in a higher percentage of acceptable trials and should result in more efficient and reliable measures during spaceflight. The results of this ground study were presented at the NASA Human Research Program Investigators' Workshops in 2024 and the Aerospace Medical Association Annual Scientific Meeting in 2024.
Bibliography Type:	Description: (Last Updated: 10/10/2025)
Abstracts for Journals and Proceedings	Carey M, Ehrenburg M, Macaulay T, Wood S. "Comparison of active and passive head impulse testing of the horizontal vestibulo-ocular reflex: exploring the feasibility of different approaches for spaceflight." Aerospace Medical Association Annual Scientific Meeting, Chicago, Illinois, May 5-9, 2024. Abstracts. Aerospace Medical Association Annual Scientific Meeting, Chicago, Illinois, May 5-9, 2024. , May-2024
Abstracts for Journals and Proceedings	Macaulay T, Wood S, Reschke M, De Dios Y, Dervay J, Makishima T, Schubert M, Shelhamer M, Kheradmand A, Clement G "Neuro-vestibular examination during and following spaceflight (vestibular health)." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Carey M, Ehrenburg M, Macaulay T, Wood S. "Comparison of active and passive head impulse testing of the horizontal vestibulo-ocular reflex: Exploring the feasibility of different approaches for spaceflight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Articles in Peer-reviewed Journals	Clément G, Moudy SC, Macaulay TR, Bishop MO, Wood SJ. "Mission-critical tasks for assessing risks from vestibular and sensorimotor adaptation during space exploration." Front Physiol. 2022 Nov 25;13:1029161. https://doi.org/10.3389/fphys.2022.1029161 ; PubMed PMID: 36505047; PubMed Central PMCID: PMC9733831 , Nov-2022
Articles in Peer-reviewed Journals	Clément G, Kuldavletova O, Macaulay TR, Wood SJ, Navarro Morales DC, Toupet M, Hautefort C, Van Nechel C, Quarck G, Denise P. "Cognitive and balance functions of astronauts after spaceflight are comparable to those of individuals with bilateral vestibulopathy." Front Neurol. 2023 Oct 27;14:1284029. https://doi.org/10.3389/fneur.2023.1284029 ; PMID: 37965165 PMCID: PMC10641777 , Oct-2023
Articles in Peer-reviewed Journals	Clément G, Kuldavletova O, Macaulay TR, Wood SJ, Navarro Morales DC, Toupet M, Hautefort C, Van Nechel C, Quarck G, Denise P. "Cognitive and balance functions of astronauts after spaceflight are comparable to those of individuals with bilateral vestibulopathy." Front Neurol. 2023 Oct 27;14:1284029. 10.3389/fneur.2023.1284029 ; PubMed PMID: 37965165; PubMed Central PMCID: PMC10641777 , Nov-2023